

Static Routing

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R1
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PFR

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
10.0.0.0/8 is directly connected, Ethernet0/0
10.0.0.1/32 is directly connected, Ethernet0/0
15.0.0.0/8 [1/0] via 20.0.0.2
20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
20.0.0.0/8 is directly connected, Ethernet0/1
20.0.0.1/32 is directly connected, Ethernet0/1
21.0.0.0/8 [1/0] via 20.0.0.2
30.0.0.0/8 [1/0] via 20.0.0.2
162.14.0.0/16 [1/0] via 20.0.0.2
172.19.0.0/16 [1/0] via 20.0.0.2
192.168.168.0/24 [1/0] via 20.0.0.2

R1#show ip interface brief
Interface IP-Address OK? Method Status Prot
Ethernet0/0 10.0.0.1 YES NVRAM up up
Ethernet0/1 20.0.0.1 YES NVRAM up up
Ethernet0/2 unassigned YES NVRAM down down
Ethernet0/3 unassigned YES NVRAM down down
R1#
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R2
R2(config)#ip route 192.168.168.0 255.255.255.0 30.0.0.2
R2(config)#ip route 162.14.0.0 255.255.0.0 30.0.0.2
R2(config)#
R2(config)#end
R2#
Aug 24 09:47:29.759: $SYS-5-CONFIG_I: Configured from console by console
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PFR

Gateway of last resort is not set

10.0.0.0/8 [1/0] via 20.0.0.1
15.0.0.0/8 [1/0] via 30.0.0.2
20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
20.0.0.0/8 is directly connected, Ethernet0/0
20.0.0.2/32 is directly connected, Ethernet0/0
21.0.0.0/8 [1/0] via 30.0.0.2
30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
30.0.0.0/8 is directly connected, Ethernet0/1
30.0.0.1/32 is directly connected, Ethernet0/1
162.14.0.0/16 [1/0] via 30.0.0.2
172.19.0.0/16 [1/0] via 30.0.0.2
192.168.168.0/24 [1/0] via 30.0.0.2

R2#write
Building configuration...
[OK]
R2#show ip interface brief
Interface IP-Address OK? Method Status Protocol
Ethernet0/0 20.0.0.2 YES manual up up
Ethernet0/1 30.0.0.1 YES manual up up
Ethernet0/2 unassigned YES TFTP down down
Ethernet0/3 unassigned YES TFTP down down
R2#write
Building configuration...
[OK]
R2#
```

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R3
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from Pfr

Gateway of last resort is not set

S 10.0.0.0/8 [1/0] via 30.0.0.1
S 15.0.0.0/8 [1/0] via 172.19.0.2
S 20.0.0.0/8 [1/0] via 30.0.0.1
S 21.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 21.0.0.0/8 is directly connected, Ethernet0/2
C 21.0.0.1/32 is directly connected, Ethernet0/2
S 30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 30.0.0.0/8 is directly connected, Ethernet0/0
C 30.0.0.2/32 is directly connected, Ethernet0/0
S 162.14.0.0/16 [1/0] via 21.0.0.2
S 172.19.0.0/16 is variably subnetted, 2 subnets, 2 masks
C 172.19.0.0/16 is directly connected, Ethernet0/1
C 172.19.0.1/32 is directly connected, Ethernet0/1
S 192.168.168.0/24 [1/0] via 21.0.0.2
R3#
R3#
R3#show ip interface brief
Interface IP-Address OK? Method Status Prot
GigabitEthernet0/0 30.0.0.2 YES manual up up
GigabitEthernet0/1 172.19.0.1 YES manual up up
GigabitEthernet0/2 21.0.0.1 YES manual up up
GigabitEthernet0/3 unassigned YES TFTP down down
R3#write
Building configuration...
[OK]
R3#
```

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R4
Gateway of last resort is not set

S 10.0.0.0/8 [1/0] via 172.19.0.1
S 15.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 15.0.0.0/8 is directly connected, Ethernet0/0
C 15.0.0.1/32 is directly connected, Ethernet0/0
S 20.0.0.0/8 [1/0] via 172.19.0.1
S 21.0.0.0/8 [1/0] via 172.19.0.1
S 30.0.0.0/8 [1/0] via 172.19.0.1
S 162.14.0.0/16 [1/0] via 172.19.0.1
S 172.19.0.0/16 is variably subnetted, 2 subnets, 2 masks
C 172.19.0.0/16 is directly connected, Ethernet0/1
C 172.19.0.2/32 is directly connected, Ethernet0/1
S 192.168.168.0/24 [1/0] via 172.19.0.1
R4#
R4#show ip interface brief
Interface IP-Address OK? Method Status Prot
GigabitEthernet0/0 15.0.0.1 YES NVRAM up up
GigabitEthernet0/1 172.19.0.2 YES NVRAM up up
GigabitEthernet0/2 unassigned YES NVRAM down down
GigabitEthernet0/3 unassigned YES NVRAM down down
R4#write
Building configuration...
[OK]
R4#
```

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R5
+ - replicated route, % - next hop override, p - overrides from PFR
Gateway of last resort is not set
S    10.0.0.0/8 [1/0] via 21.0.0.1
S    15.0.0.0/8 [1/0] via 21.0.0.1
S    20.0.0.0/8 [1/0] via 21.0.0.1
S    21.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    21.0.0.0/8 is directly connected, Ethernet0/0
L    21.0.0.2/32 is directly connected, Ethernet0/0
S    30.0.0.0/8 [1/0] via 21.0.0.1
S    162.14.0.0/16 [1/0] via 192.168.168.2
S    172.19.0.0/16 [1/0] via 21.0.0.1
S    192.168.168.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.168.0/24 is directly connected, Ethernet0/1
L    192.168.168.1/32 is directly connected, Ethernet0/1
R5#
R5#
R5#show ip interface brief
Interface              IP-Address      OK? Method Status      Prot
lo0                    10.0.0.1        YES NVRAM  up          up
Ethernet0/0            21.0.0.2        YES NVRAM  up          up
Ethernet0/1            192.168.168.1   YES NVRAM  up          up
Ethernet0/2            unassigned      YES NVRAM  down        down
Ethernet0/3            unassigned      YES NVRAM  down        down
R5#write
Building configuration...
[OK]
R5#

```

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R6
Gateway of last resort is not set
S    10.0.0.0/8 [1/0] via 192.168.168.1
S    15.0.0.0/8 [1/0] via 192.168.168.1
S    20.0.0.0/8 [1/0] via 192.168.168.1
S    21.0.0.0/8 [1/0] via 192.168.168.1
S    30.0.0.0/8 [1/0] via 192.168.168.1
S    162.14.0.0/16 is variably subnetted, 2 subnets, 2 masks
C    162.14.0.0/16 is directly connected, Ethernet0/0
L    162.14.0.1/32 is directly connected, Ethernet0/0
S    172.19.0.0/16 [1/0] via 192.168.168.1
S    192.168.168.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.168.0/24 is directly connected, Ethernet0/1
L    192.168.168.2/32 is directly connected, Ethernet0/1
R6#
R6#
R6#
R6#show ip interface brief
Interface              IP-Address      OK? Method Status      Prot
lo0                    10.0.0.1        YES NVRAM  up          up
Ethernet0/0            162.14.0.1      YES NVRAM  up          up
Ethernet0/1            192.168.168.2   YES NVRAM  up          up
Ethernet0/2            unassigned      YES NVRAM  down        down
Ethernet0/3            unassigned      YES NVRAM  down        down
R6#write
Building configuration...
[OK]
R6#

```



